CODERED '26 | INDUSTRY CHALLENGE BRIEF

Problem Statement 3
Catalog Genius

AI-Powered Product Generation & Authenticity Detection

OBJECTIVE Simplify e-commerce product listing while ensuring the authenticity and reliability of product images, using generative AI and synthetic media detection.

1. Challenge Overview

Company / Partner Name: MyStartupWave

Challenge Title: Catalog Genius – AI-Powered Product Generation & Authenticity Detection

Industry / Track: E-commerce / Generative AI & Digital Trust

Challenge Contact: MyStartupWave Challenge Desk (mystartupwave@gmail.com)

Challenge Type: Product Innovation

Brief Summary: Teams will develop an AI-powered platform that generates high-quality product images and detailed e-commerce listings from minimal text descriptions or uploaded product photos. The same platform must detect whether uploaded images are authentic photographs or AI-generated, with confidence scores and explainable indicators, helping vendors maintain transparency and trust in online marketplaces.

2. Industry Context

India's e-commerce and social-commerce ecosystem is growing rapidly, with millions of small sellers, D2C brands and ONDC-enabled merchants coming online. Every listing needs clean images on a white background, multiple angles, a compelling title, a structured description, specifications and SEO tags, and every marketplace has its own image and content rules. Professional photography and copywriting are expensive and slow, so many vendors upload poor-quality photos and incomplete listings, which lowers conversion and increases returns.

At the same time, image-generation models can now produce photorealistic product images in seconds. This can democratise catalog creation, but it also creates a new risk: AI-generated images that misrepresent what the buyer will actually receive. Marketplaces, consumers and regulators are paying increasing attention to transparency in synthetic media, creating demand for tools that combine fast catalog creation with trustworthy authenticity analysis.

3. Problem Statement

- **Problem and who is affected:** Small and medium sellers, D2C brands and catalog teams must create large numbers of listings quickly and cheaply, while buyers need to trust that product images reflect the real item.
- **When and where it occurs:** During onboarding of new products, seasonal catalog refreshes and expansion to new marketplaces, and at the moment a buyer decides whether an image can be trusted.
- **Limitations of current solutions:** Studios and freelance copywriters are costly and slow; generic AI image tools ignore marketplace guidelines and often alter the product itself (wrong logos, colours, shapes or text); listing generators produce generic copy; and AI-image detectors are mostly standalone, unreliable on compressed or edited images, and give a bare "real/fake" answer with no explanation.

- **Why it is worth solving:** Image quality and listing completeness are widely recognised drivers of conversion and return rates; marketplaces publish strict image and content guidelines; and concern about synthetic media and misleading product imagery is growing among platforms, consumers and policymakers.

4. Target Users and Stakeholders

Primary users: Small and medium online sellers, D2C brands and independent vendors selling on marketplaces and on Shopify or WooCommerce stores.

Secondary users / stakeholders: Marketplace catalog and trust & safety teams, e-commerce agencies, and end buyers.

Current user journey or workflow: Take product photos on a phone → edit them manually or pay for photography → write titles and descriptions by hand → guess SEO keywords → upload to each marketplace separately → fix rejections for guideline violations. Image authenticity is rarely checked at all.

Key user pain points: High cost and time per listing; inconsistent quality across products; repeated rework for different platforms; and no easy way to prove images are genuine or to detect AI-generated images supplied by third parties.

5. Product Opportunity

- **What users should accomplish:** Go from a short text description or a single phone photo to a complete, marketplace-ready listing (multiple professional images plus title, description, specifications, tags and SEO content) in minutes.
- **Meaningful improvement:** Generated images that stay faithful to the real product, so listings are attractive without being misleading, and an authenticity analysis that gives vendors and marketplaces a confidence score with human-understandable evidence.
- **What makes it different:** A single platform that unites catalog creation and trust verification, so speed and transparency reinforce each other instead of being traded off.

6. Expected Product Capabilities

Must-have capabilities:

- AI Product Image Generation: professional, high-quality product images from text descriptions or reference photos, including clean white-background packshots and at least one lifestyle/context scene.
- Product fidelity: when a reference photo is supplied, generated and enhanced images must preserve the product's shape, colour, branding and printed text, and the system must flag outputs that fail its fidelity checks.
- Smart Image Enhancement: background removal, lighting and colour correction, upscaling, and generation of multiple consistent product views (e.g., front, side, angled).
- AI-Powered Product Listings: automatically generated titles, highlights, descriptions, structured specifications, tags/categories and SEO-friendly content, adapted to the character limits and rules of at least two target platforms.
- Image Authenticity Detection: analyse uploaded images and estimate whether each is a real photograph or AI-generated, returning a calibrated confidence score.
- Trust & Transparency: an explainable report for each verdict with visual indicators (e.g., heatmaps of suspicious regions), metadata and provenance checks (EXIF, C2PA Content Credentials where present) and plain-language reasons.
- E-commerce Integration: one-click export to Shopify and WooCommerce, via their APIs or official product CSV import formats.

Nice-to-have capabilities:

- Batch processing of an entire catalog (e.g., 50+ products from a CSV file).

- Multilingual listings (English plus at least one Indian language).
- Embedding disclosure labels or C2PA Content Credentials in generated images so they are transparently marked as AI-generated.
- Detection of partial manipulation (splicing, object insertion or removal) in addition to fully synthetic images.
- A marketplace-guideline compliance checker (image dimensions, background, text overlays).

Expected user interactions: The vendor enters a short description or uploads a phone photo → reviews and edits the generated images and listing → runs an authenticity check on any image → reviews the trust report → exports to Shopify or WooCommerce, or downloads a complete listing pack.

Expected demonstration / deliverable:

- Live creation of at least three complete listings across different categories (e.g., apparel, an electronics accessory, packaged food) from minimal input.
- Live authenticity analysis on a judge-provided mixed set of real and AI-generated images, including compressed and cropped versions.
- Successful export to a Shopify development store or a WooCommerce test site.

7. Technical and Operational Constraints

Technical or platform constraints: Web-based platform; any model stack may be used (open-source or API). Authenticity detection must not rely on metadata alone, because metadata is easily stripped: it must include content-based analysis of the pixels. Exports must follow the official Shopify and WooCommerce product formats.

Performance, accuracy, or latency requirements: A complete listing (images and text) in under 60 seconds per product. Authenticity detection accuracy of at least 85% on the judges' balanced hidden test set, with no more than a 10-percentage-point drop on JPEG-compressed, resized or cropped versions of the same images. Detection latency under 5 seconds per image. Generated text must not contain specifications that are absent from the input.

Security, privacy, or compliance requirements: Uploaded images and data are used only for the requested task. No use of copyrighted or third-party brand imagery without permission. Generated images must be disclosed as AI-generated in exported listings, in line with the principles of India's Consumer Protection (E-Commerce) Rules, 2020 against misleading product information.

Hardware, connectivity, or deployment limitations: Should run on cloud or free GPU tiers. The vendor-facing interface must work on mobile browsers, since many vendors operate from their phones.

Budget or resource constraints: Teams must document the generation cost per listing. The solution should be economically viable for small vendors, with a target cost well below that of a single professional product photograph.

Domain-specific safety requirements: Authenticity results are probabilistic and must be presented as estimates with confidence levels, never as definitive proof. To avoid false accusations, uncertainty must be communicated clearly, including an "inconclusive" band.

8. Available Resources and Support

Resources marked "Confirmed" are guaranteed by the partner. "Permitted" items may be used by teams at their own discretion and cost. "Under consideration" items are not yet confirmed.

APIs, SDKs, developer tools, or technology credits: Confirmed: none provided by the partner. Permitted: open-source image-generation and vision models, or generative AI APIs available to the team. Shopify development stores (via the Shopify Partner programme) and WooCommerce can be set up free of charge.

Sample datasets or permitted data sources: Self-captured product photos, public-domain or openly licensed product images, and public real-vs-synthetic image datasets used within their licences. Judges will evaluate detection on a hidden test set.

Sandbox environments or test infrastructure: A Shopify development store and a local or hosted WooCommerce test site, set up by each team.

Documentation and reference materials: Shopify and WooCommerce product API/import documentation; the C2PA (Content Credentials) specification; major marketplaces' image guidelines; Consumer Protection (E-Commerce) Rules, 2020.

Mentors, domain experts, or technical sessions: Confirmed: MyStartupWave engineers during the scheduled mentoring rounds (subject to the event schedule).

Other resources or support: Under consideration: a session on synthetic media detection and responsible disclosure.

9. Business and Adoption Context

Intended customer or beneficiary: Small sellers and D2C brands (SMB SaaS); marketplaces and e-commerce agencies (B2B API).

Current alternatives or competing solutions: Professional photographers and studios, AI background-removal and product-shot tools, general-purpose LLM copywriting, and standalone AI-image detectors.

Potential value created: Lower cost and time per listing, faster catalog launches, higher conversion from better images and copy, fewer returns caused by misleading images, and stronger buyer trust.

Potential adoption pathway: Freemium, per-listing pricing for SMBs; a Shopify / WooCommerce plugin; an API for marketplace trust & safety teams.

Commercial or operational considerations: GPU inference costs, changes in marketplace policies, maintenance of platform integrations, and liability considerations around authenticity verdicts.

Known assumptions or uncertainties: Detectors may not generalise to future image-generation models; vendors are assumed to accept AI-generated images when they stay faithful to the real product and are clearly disclosed.

10. Success Indicators

Functional outcomes: End-to-end listing generation and export working for all demo products; an authenticity report produced for every test image.

User or customer outcomes: Time to create a complete listing reduced from hours to under 5 minutes; vendors rate generated images as publishable in at least 80% of cases.

Technical performance indicators: ≥ 85% detection accuracy on clean images and ≤ 10-point drop under common transformations; well-calibrated confidence scores; < 60 seconds per listing; product fidelity preserved in judge review.

Business or operational indicators: Cost per listing and a pricing model clearly justified for small sellers.

Safety, reliability, or compliance indicators: No hallucinated product specifications; clear disclosure of AI-generated imagery; uncertainty communicated for every authenticity verdict.

11. Scope and Boundaries

In scope: Image generation and enhancement, listing generation, authenticity detection with explanations, and Shopify / WooCommerce export.

Out of scope: Payment processing, inventory and order management, live publishing on production marketplaces, legal certification of authenticity, and video generation.

Permitted assumptions or simplifications: Teams may focus on 2–3 product categories. Shopify and WooCommerce integration may use test stores. Detection may target current mainstream image generators.

Dependencies (systems, approvals, resources): Access to image-generation and vision models, GPU compute (free tiers acceptable) and test e-commerce stores.

12. Industry Engagement

Will the company provide mentors or subject-matter experts? Yes. Engineers from MyStartupWave will be available during the scheduled mentoring rounds for architecture, product and feasibility reviews.

Will teams receive feedback during development? Yes. Teams will receive structured feedback at the designated mid-event checkpoint(s), focused on scope, technical approach and progress against the success indicators.

Will the company participate in evaluation or demonstrations? Yes. MyStartupWave representatives will attend the final demonstrations and take part in evaluation alongside the CODERED ’26 jury.

Is further validation, pilot discussion, or product exploration possible after the event? Possible. Teams with exceptional solutions may be invited to a follow-up discussion on further validation, mentoring or pilot exploration.

Any follow-on engagement is subject to mutual interest and separate agreement.

13. Intellectual Property and Participation Terms

The following terms apply to all participating teams and are communicated before development begins. Participation does not imply any automatic transfer of ownership or commercial rights; any subsequent arrangement must be agreed separately.

Data and resource usage conditions: Use only self-captured, public-domain or properly licensed images and datasets, within their licence terms. Do not scrape product images or listings from marketplaces in violation of their terms of service.

Confidentiality requirements, if any: None. This brief and any materials shared for this challenge are non-confidential.

Intellectual property and ownership terms: Teams retain full ownership of the code, designs, models and other intellectual property they create. Participation does not transfer any ownership or commercial rights to MyStartupWave.

Restrictions on using company names, data, or materials: Do not use MyStartupWave’s name or branding without written permission. Third-party brands, logos and marketplace names may be referenced only descriptively and must not be used to generate misleading product imagery.

Terms governing subsequent pilots, licensing, investment, employment, or commercial collaboration: Any pilot, licensing, investment, employment or commercial collaboration after the event is subject to mutual interest and a separate written agreement.

14. Partner Contact and Clarifications

Primary contact: MyStartupWave Challenge Desk

Email: mystartupwave@gmail.com

Reference documentation / resource links: This challenge brief; Shopify and WooCommerce developer documentation; C2PA specification (c2pa.org); Consumer Protection (E-Commerce) Rules, 2020.

Clarification process: Send questions in writing to the challenge contact or raise them through the official CODERED ’26 helpdesk. Answers that affect scope or evaluation will be shared with all teams on this challenge at the same time, to keep the competition fair.

Submission Checklist

- ☑ The problem is real, relevant, and clearly stated.
- ☑ Target users and desired outcomes are identified.
- ☑ Teams have freedom to develop their own solutions.
- ☑ A working prototype is reasonably achievable within the challenge timeline.
- ☑ Business value and adoption can be evaluated.
- ☑ Technical constraints and available resources are stated.
- ☑ Data, safety, confidentiality, and IP terms are clear.
- ☑ Evaluation expectations are communicated consistently.

THE GUIDING PRINCIPLE

The company defines the opportunity. The team defines the solution.